

Fraunhofer FKIE Drones

Researchers at Germany's Fraunhofer FKIE research lab are turning drones into life-savers by kitting them out with additional features and capabilities. This includes microphone arrays, smart AI algorithms, and an onboard computer. With these additions, the drone can detect the sounds of screams and pinpoint their location. As of right now, the research lab uses kitted DJI Matrice M600 drones for proof-of-concept tasks. According to the research lab, any UAV could easily be swapped out with the one currently being used, as long as they can carry the requisite payload of 1kg.



According to acarena Varela, a research fellow at Fraunhofer FKIE who worked on the project, the prime use case for these drones would be after disasters, such as earthquakes for example. In these situations, every minute is important. To those victims trapped under rubble or who are injured, those 60 seconds are crucial. Drones are already used in disaster situations thanks to their ability to survey from the skies and their ability to reach difficult places. But for 8 years now, Varela and Fraunhofer FKIE have been working to make them capable of more in these situations and it's starting to bear fruit.



MIT Digital Fiber

Researchers at MIT have created what they're calling the world's first fiber with digital capabilities. This fiber can sense, store, analyse and infer activity, even after being sewn into a clothing item. Yoel Fink, a professor of material sciences and electrical engineering, and senior author on the study, said that "digital fibers expand the possibilities for fabrics to uncover the context of hidden patterns in the human body that could be used for physical performance monitoring, medical inference, and early disease detection". Electronic fibers have been analogue until now, with a steady flow of electrical signals. With digital fiber, information can now be stored and processed, in the form of 1s and 0s. This adds a whole new dimension to textiles and fabrics, wherein they could be programmed literally. The new fiber was created by placing hundreds of square silicon microscale digital chips into a preform that was then used to create a polymer fiber. It's thin enough to be passed through a needle, and you won't feel it at all when you're wearing it, according to MIT PhD student Gabriel Loke and MIT postdoc Tural Khudiyev, the lead authors on the paper. With these new fibers, the possibilities are limitless, from digitally turning on lights, to unlocking doors, and even storing information directly on your clothes.

Tamagotchi Smart

For its 25th anniversary, the Tamagotchi is making a comeback, as a wearable. You'll be able to wear your digital pet on your wrist, and interact with it via touch and your voice.

The original Tamagotchi debuted in 1996, from Bandai. The new Tamagotchi smartwatch features a touchscreen LCD and a battery life of 30 hours. It can be charged via USB. With it being worn around your wrist like a smartwatch and all, it certainly has some additional features, including being able to tell the time, of course. All the better for you to remember to feed your pet on time. The Tama-



gotchi Smart will not be going on sale directly for now, but will instead be sold by lottery. Unfortunately, the time window to apply for the lottery has already passed. The winners, however,

are expected to be announced sometime later this month. The wearable is expected to go on sale sometime late in November, and will retail for approximately \$68. Additionally, you can also buy "smart cards" which can be inserted into the device for \$10 each. These cards will give you themed items and food which you can feed your pet, as well as additional mini-games and clock faces in some cases.